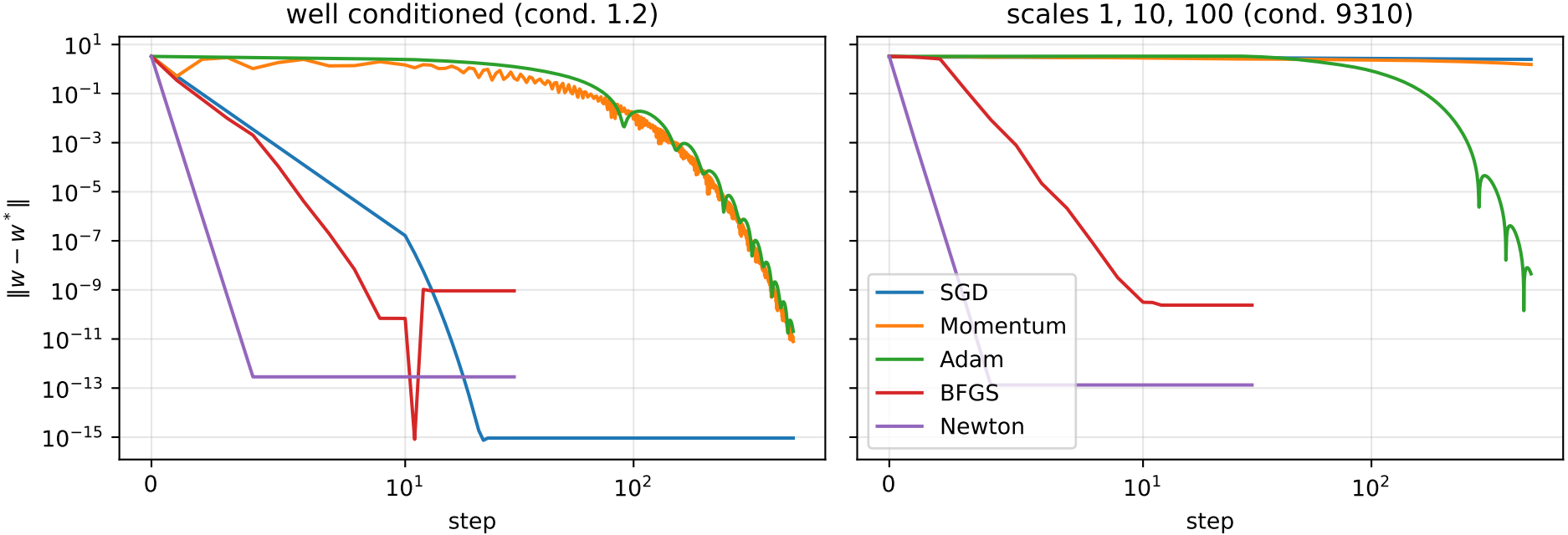
well conditioned (cond. 1.2)
scales 1, 10, 100 (cond. 9310)
$\|w - w^*\|$
10^{1}
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{-9}
10^{-11}
10^{-13}
10^{-15}
0
10^{1}
10^{2}
step
SGD
Momentum
Adam
BFGS
Newton